

IBDC Protocol - Sequence diagram

Embedded system
(IBDC device)

BLE Link

Phone
(APP)

Event
Detected

(NOTIFY)

EventNotification (event-id,
distance-cm, time-offset-ms, image-count,
image-format)

Phone
Request
Image
Transfer

Image Transfer Request (
event-id, start-from-image,
start-from-chunk)

(NOTIFY)

Device Sends
Image
Info

(NOTIFY)

NEW

Image Info (repeated per image)
(event-id, image-index, image-size-bytes,
total-chunks, image-format)

Device
sends
Image
chunks

Image Chunk (event-id, image-index
chunk-sequence=0, total-chunks, payload)
is-last-chunk

Image Chunk (" ", chunk-sequence=N
is-last-chunk=true)

Phone
Ack
Event

Event Transfer Ack
NEW (event-id)

NOTE: Device may
delete images from
storage after
Ack.

A.

Missed
(or late
Retrieval)
Notifications

Pending Event List Request

Pending Event List
(event-id)

B.

Request
Info
for specific
Event

Event Info Request
(event-id)

Event Notification (
event-id, distance-cm, time-offset-ms,
image-count, image-format)

Repeat Normal case